March 8, 1960 J. KAYE ET AL 2,927,398

MULTIPLE STAGE ROCKET

Filed May 13, 1958 2 Sheets-Sheet 1

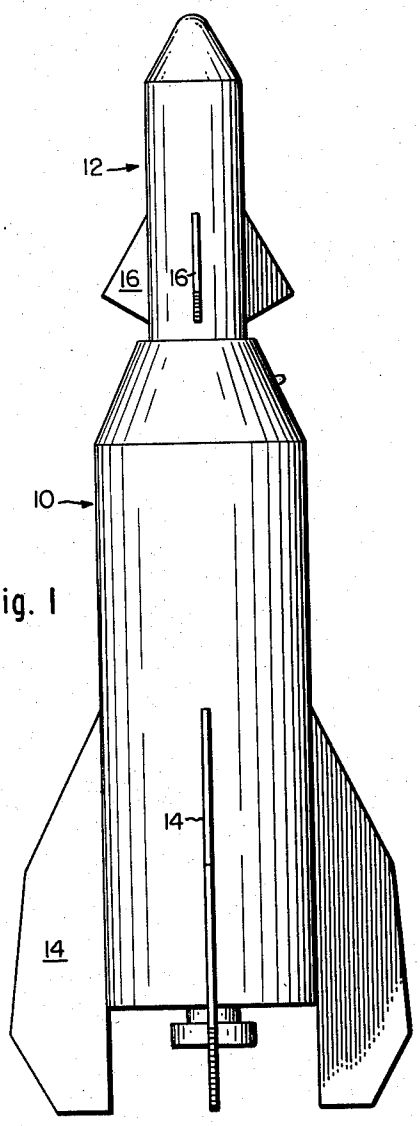


Fig. 1

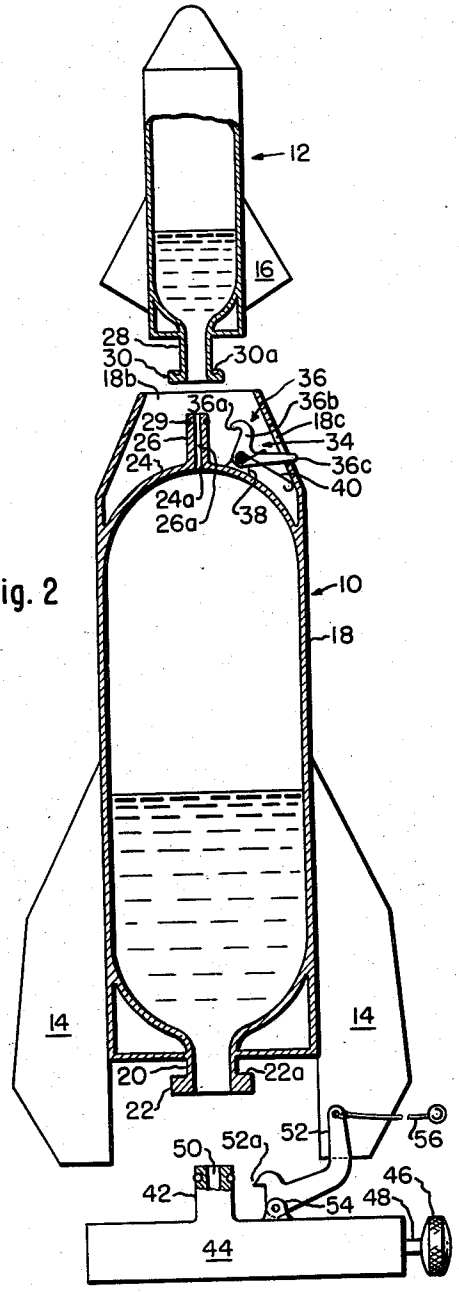


Fig. 2

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Fig. 3

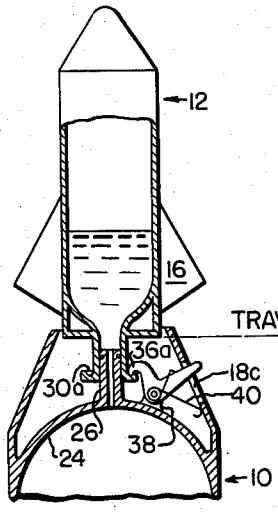


Fig. 4

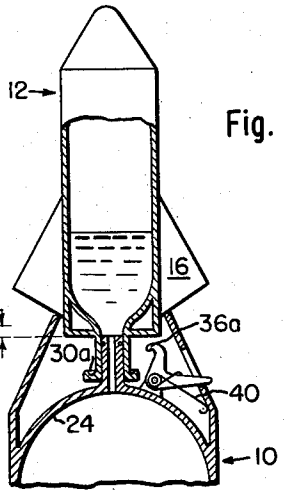


Fig. 5a

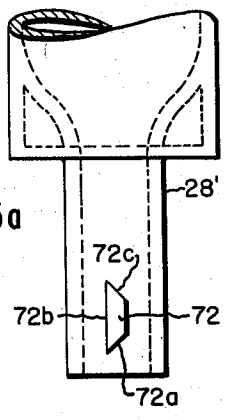


Fig. 5b

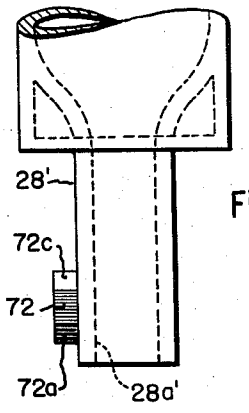


Fig. 6

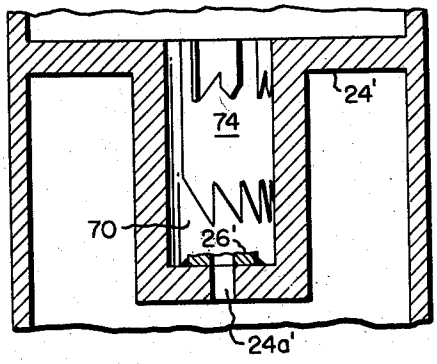
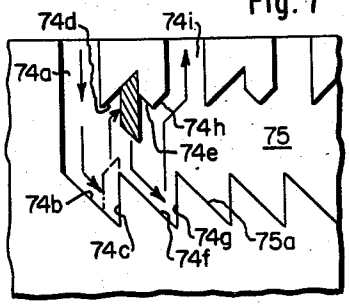


Fig. 7



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MULTIPLE STAGE ROCKET

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Application May 13, 1958, Serial No. 735,016

19 Claims. (Cl. 46—74)

Our invention relates to multiple stage rockets. More particularly it relates to a two-stage rocket propelled by noncombustible propellants. Because of the utilization of noncombustible propellants, rockets of the general type herein described are particularly useful as toys. However, rockets of this type may also be used for other purposes, such as the safe and inexpensive testing of equipment for use on rockets of greater range and power than toy rockets. Accordingly, while our invention will be described as applied to a realistic two-stage toy rocket, it will be understood that it may also be applied to rockets used for many other purposes, including rockets using combustible fuels.

Heretofore rockets using noncombustible propellants have been of the single stage type. In general they comprise a rocket shaped vessel, usually substantially cylindrical in shape in the central portion, with a nose section shaped as an ogive. The tail section of the rocket tapers to a nozzle or orifice through which fluid under pressure is expelled to provide thrust. Stabilizing fins are usually attached to the tail section to insure a reasonably stable flight.

In use the interior of these rockets may be partially filled with a liquid, which may be water, when the rocket nose is in the downward position. The rocket is then turned so that the nose is pointed upwardly and the space defined by the upper surface of the liquid and the upper walls of the rocket is pressurized. During pressurization the orifice in the tail section is sealed and the rocket is restrained to the launching mechanism. When it is desired to launch the rocket, the restraint is removed and the thrust resulting from the expulsion of fluid from the nozzle in the tail section propels the rocket in the desired direction.

It is well known that multiple stage rockets can achieve greater velocities and higher trajectories than the single stage type heretofore described. In rockets utilizing combustible propellants, the second stage is arranged to fire after the thrust of the first stage has been expended. Means are also usually provided, either in conjunction with the firing of the rocket motor, or separately, for blowing the second stage away from the first with a controlled explosion. However, this is not desirable in a safe rocket utilizing noncombustible propellants. Also, any mechanism for controlling the time of firing of the second stage is extremely complex and expensive, and not adapted for use in rockets designed as toys. Thus, there has not heretofore been available a multiple stage rocket utilizing a noncombustible propellant. As a toy such a rocket would have appeal as being more nearly like the rockets currently in use for military, scientific and satellite purposes. For other purposes, such as equipment testing, the higher velocities and trajectories which could be achieved with a multiple stage rocket would more nearly simulate conditions which might be encountered in a rocket using combustible propellants.

Accordingly, it is a principal object of our invention

to provide a multiple stage rocket utilizing noncombustible propulsion means. Another important object of our invention is to provide multiple stage rockets utilizing nonexplosive means for separation of the stages thereof.

Another object of our invention is to provide a rocket of the type described in which release of each succeeding stage takes place after the thrust of the prior stage has been substantially diminished in order to take maximum advantage of the thrust of the prior stage.

A further object of our invention is to povide a rocket of the type described having a release mechanism which will maintain the two stages of said rocket in engagement during pressurization, but which will permit the second and subsequent stages to separate from the prior stage at the appropriate time.

Still another object of our invention is to provide a rocket of the type described wherein the weights and operating pressures of each stage are so chosen, with respect to each other, that the second and additional stages will remain in contact with the first stage for an appreciable time after the rocket is launched, without the provision of special apparatus for holding them in engagement.

A still further object of our invention is to provide a rocket of the type described which is simple, safe and reliable in operation, and rugged in construction so that it may conveniently be embodied in a toy rocket or reusable test equipment.

Yet another object of our invention is to provide a rocket of the type described which is economical of manufacture. Other and further objects of our invention will in part be obvious and will in part appear hereinafter.

The invention accordingly comprises the features of construction, combinations of elements, and arrangements of parts which will be exemplified in the constructions hereinafter set forth, and the scope of the invention will be indicated in the claims.

For a fuller understanding of the nature and objects of our invention, reference should be had to the following detailed description taken in connection with the accompanying drawings in which:

Fig. 1 is a pictorial representation of the exterior of a two-stage rocket made according to our invention;

Fig. 2 is a fragmentary sectional view of the rocket and the launcher associated therewith with the two rocket sections separated, and illustrating one type of release mechanism;

Fig. 3 is a fragmentary longitudinal section showing the manner in which the first and second stages of the rocket are engaged during pressurization, using the release mechanism of Fig. 2;

Fig. 4 is a view similar to Fig. 3 showing the relative positions of the first and second stages immediately after release of the holding means.

Fig. 5*a* is a fragmentary plan view of the nozzle end of the second stage of a rocket utilizing a second type of release mechanism;

Fig. 5*b* is a fragmentary side elevation of the nozzle portion of the second stage shown in Fig. 5*a*;

Fig. 6 is a fragmentary section of the upper end of the first stage illustrating the construction of this stage when a release mechanism of the second type is used, and

Fig. 7 is a detail of the slideway of Fig. 6.

In general, two-stage rockets using noncombustible propellants made according to our invention include a relatively large first stage, generally indicated at **10** in Fig. 1 and a substantially smaller second stage, generally indicated at **12**. Both stages are provided with the usual stabilizing fins, indicated at **14** and **16** respectively.

Each stage is hollow, so that pressurized fluid may be placed therein and includes a nozzle at the lower end

for discharge of the propellant. Prior to launching of the rocket we provide means for holding the second stage to the nose portion of the first stage. The second stage may be sealed by the holding means if desired, but we prefer to allow a passage to exist between the first and second stages as will be explained more fully hereinafter. The holding means permits both stages to be pressurized, but prevents them from separating prior to launch. To accomplish this holding and separation, a slight amount of relative longitudinal travel is permitted between the first and second stages. The difference in pressure between the inside and the outside of the second stage produces a relative force which holds the second stage at one end of this travel against the holding means. However, when the first stage is launched, the inertia of the second stage causes it to move to the other end of this permitted travel. In moving from one end of this permitted travel to the other, the holding means is uncoupled, and the second stage is free to push away from the first stage when the relative force previously mentioned is sufficient to overcome the inertial force created by the first stage which tends to hold the second stage against its nose.

As previously mentioned, we provide a passage between the first and second stages for simplicity of pressurization. Utilizing a rocket whose interior is partially filled with liquid, both first and second stages may be pressurized simultaneously. This construction, because of its extreme simplicity, makes multiple stage rockets which are made according to our invention very economical of manufacture.

More specifically, as shown in Fig. 2, the first stage **10**, includes an elongated cigar-shaped body **18** having a nozzle **20** formed at its lower end. A flange **22** is provided at the lower end of the nozzle for purposes to be hereinafter described. A bulkhead **24** is provided in the interior of the body **18** near the upper end thereof and a small opening **24***a* is formed in this bulkhead. An upstanding tube **26** is affixed to the upper surface of the bulkhead, the bore **26***a* of the tube registering with the opening **24***a* in the bulkhead. The body **18** of the rocket, the bulkhead **24** and tube **26** may be made of metal or a plastic having sufficient tensile strength to withstand the internal pressures within the closed chamber formed by the body **18** and the bulkhead **24**. An opening **18***b* is formed in the upper end of the body **18** for the coupling of the second stage.

In Fig. 2 it will be observed that the nozzle **28** of the second stage is elongated and has an internal diameter sufficient to slip over the outside of the tube **26**. An O-ring **29** or other conventional sealing device is secured to the tube **26** to insure a fluid-tight seal between the second stage nozzle and the tube when the second stage is coupled thereto. The remainder of the second stage is also an elongated hollow fluid-tight body, whose only opening is the nozzle **28**. If more than two stages are used, the nose of the second stage could of course be constructed in the same or similar fashion to that of the first stage.

A flange **30** is formed at the lower end of the nozzle **28** having an annular groove **30***a* formed on the upper surface thereof. This groove, together with the holding means generally indicated at **34** serves to hold the two stages of the rocket together during pressurization. In the embodiment illustrated in Figs. 2, 3 and 4, the holding means includes an L-shaped rod **36** having a hook **36***a* formed on the upper end of the vertical leg **36***b* of the L. The rod is pivoted in the bracket **38**, which is fixed to the bulkhead **24**, the pivot point being close to the bend in the L. In Fig. 2, we have shown the rod as being pivoted on its horizontal leg, but it will be obvious that it might also be pivoted on the vertical leg **36***b*.

The horizontal leg **36***c* extends outwardly through an opening **18***c* formed in the side of the body **18** above the bulkhead **24**, where it may readily be grasped by the operator. A spring **40** or other resilient means is connected between the horizontal leg **36***c* and the bulkhead to bias the rod **36** in the position illustrated in Figs. 2 and 4.

To operate the rocket, the first stage is held nose downward and the amount of water or other fluid which it is desired to use in both the first and second stages is fed into the first stage through the nozzle **20**. The water does not flow out the opening **24***a* and tube **26**, while the nozzle **20** is sealed, since the bore of tube **26** is preferably of a sufficiently small diameter so that the water or other fluid utilized to propel the rocket will not flow through it. For example, because of capillary and surface tension effects, water will not flow out of an otherwise sealed container through a tube having a diameter much less than ⅜ of an inch when the tube is made of glass or of certain plastic materials.

After the water is placed in the interior chamber and while the first stage is in the inverted position, the nozzle **28** of the second stage is slid upwardly around the tube **26** of the first stage. The operator then grasps the arm **36***c* and causes the rod **36** to pivot so that the hook **36***a* engages the groove **30***a* in the flange **30** formed on the base of the second stage, thus holding the second stage in position against the nose portion of the first stage.

A pump, or other pressure generating device, generally indicated at **42** in Fig. 2 is then attached to the first stage nozzle. As illustrated, the pump includes a body **44** having the usual piston therein. A handle **46** is attached to the piston rod **48** in conventional fashion. An outlet **50**, whose outer diameter is substantially the same as the inner diameter of the nozzle **20** is formed on the pump, and an O-ring or other conventional seal is secured to the outer surface of the outlet tube to insure a fluid-tight seal between the pump and nozzle. An L-shaped fitting **52** is pivotally attached to the pump body as by a bracket **54**. The catch **52***a* formed on fitting **52** engages the upper surface **22***a* of the flange **22** to restrain the pressurized rocket prior to launch.

A lanyard **56** is attached to the fitting **52** to pull it out of the engagement with the flange **22** when the rocket is ready for launching.

After attachment of the pump or other pressurizing mechanism to the first stage, and with the first and second stages pointing downwardly, the pump is operated to pump fluid from the first into the second stage. When sufficient fluid is in the second stage, both stages are turned to the "nose-up" position; continued pumping provides pressure for propulsion. Because the interiors of both stages are connected by the tube **26**, the pressures in the interior of both the first and second stages will be substantially equalized. It will be understood, of course, that the two stages may be separately pressurized, in which case, the pressures in the interiors of the two stages will not necessarily be equal.

When the first stage is restrained, the second stage will exert a relative force with respect to the restrained first stage because of its internal pressure as previously described. This force causes the second stage to push away from the first stage and locks the hook **36***a* in the annular groove **30***a* as shown in Fig. 3. Also, as shown in the same figure, the spring **40** is extended and pulling on arm **36***b*. Note that the second stage can proceed farther down the tube **26** before it reaches the limit of its travel.

The appearance of the rocket immediately after launch is shown in Fig. 4. As illustrated therein, the second stage, because of its inertia, has slid down the tube **26** until its guiding fins **16** have engaged the nose of the first stage. The hook **36***a* having been freed from the annular groove **30***a* has been retracted by the action of the spring **40**. It will be obvious that other means than the engagement of the second stage fins by the nose of

the first stage may be provided to limit the downward travel of the second stage. For example, the end of the second stage nozzle might be arranged to engage bulkhead **24**.

Thus the second stage is now no longer restrained relative to the first stage. However, the second stage will not separate immediately because the acceleration of the entire assembly forces the second stage against the first stage. As the fluid continues to be expelled from the first stage, the pressure therein drops resulting in a decreasing thrust from the first stage. When the acceleration of the entire rocket decreases sufficiently due to diminution of the thrust from the first stage, the second stage separates therefrom.

It is apparent that the hold and release mechanism herein described will operate for a two-stage rocket where the separate stages are separately pressurized and no passage is provided between the two stages. It is also apparent that a third stage could be added to the construction herein described by merely providing the second stage with an opening in the nose thereof and a release mechanism similar to that of the first stage. The third stage, and subsequent stages, may be separately pressurized if desired.

We have found that to obtain a multiple stage rocket in which at least two stages remain in contact for some time after uncoupling as a result of the acceleration of the rocket, and then separate as the acceleration of the first stage diminishes, certain relationships must be maintained between the nozzle areas, the pressures and the weights of the two stages which are to remain in contact. These relationships may be determined as follows: If p_2 is the gage pressure in the second stage at the time of uncoupling and A_2 is its nozzle area, then the relative pressure force tending to separate the two stages prior to their actual separation is p_2A_2.

After launch this relative pressure force must be less than the force resulting from rocket acceleration and gravity acting on the mass M_2 of the second stage if the two stages are to remain in contact, i.e.:

$$p_2A_2<M_2\ddot{y}+M_2g \tag{1}$$

or

$$p_2A_2-M_2g<M_2\ddot{y} \tag{1a}$$

where $\ddot{y}$ is the acceleration of the first stage and g is the gravitational acceleration.

The thrust T of the first stage at the time of uncoupling equals the total mass of both stages multiplied by their acceleration, i.e.:

$$T=(M_1+M_2)\ddot{y}+(M_1+M_2)g \tag{2}$$

where M_1 is the mass of the first stage at the time the stages are uncoupled.

Solving Equation 2 for $\ddot{y}$ yields:

$$\ddot{y}=\left[\frac{T}{(M_1+M_2)g}-1\right]g \tag{2a}$$

Substituting Equation 2*a* in 1*a*,

$$p_2A_2-M_2g<M_2g\left[\frac{T}{W_1+W_2}-1\right] \tag{3}$$

W_1 and W_2 being equal respectively to M_1g and M_2g and being the values of the weight of each stage at the time of uncoupling.

Equation 3 may be further simplified to:

$$p_2A_2<\left(\frac{W_2}{W_1+W_2}\right)T \tag{3a}$$

which indicates that for the two stages to remain in contact after they are otherwise free to separate, the relative pressure force tending to separate them, p_2A_2, must be less than the ratio of the second stage weight to the total rocket weight multiplied by the thrust, T. The thrust, $T=m_1V_j$ where m_1 is the mass flow rate from the first stage and V_j is the average velocity of the first stage jet over the cross section. It is known that $\dot{m}_1=\rho V_jA_j$, where ρ is the fluid density in the first stage and A_j is the area of the jet being emitted from the first stage. A_j is related to the nozzle area of the first stage A_1 by a nozzle coefficient C which depends upon nozzle shape. Thus $A_j=CA_1$. The nozzle coefficient C is defined and representative values are given for various nozzle configurations on page 157 of Hunsaker and Rightmire, "Engineering Applications of Fluid Mechanics" (1947).

Thus, the thrust T at the time of uncoupling may be determined as

$$T=(\rho V_jA_j)V_j=\rho CA_1V_j^2 \tag{4}$$

It is well known in fluid mechanics that

$$(V_j)^2=2\left(\frac{p_1}{\rho}\right) \tag{5}$$

where p_1 is the pressure in the first stage at the time of uncoupling.

Substituting Equation 5 in 4,

$$T=2CA_1p_1 \tag{6}$$

If Equation 6 is now substituted in Equation 3*a*

$$p_2A_2<\left(\frac{W_2}{W_1+W_2}\right)2CA_1p_1 \tag{7}$$

and rearranging terms, we have

$$\frac{A_2}{A_1}<2C\left(\frac{W_2}{W_1+W_2}\right)\left(\frac{p_1}{p_2}\right) \tag{8}$$

Equation 8 defines the necessary conditions for the second and any additional stages of a multiple stage rocket made according to our invention to remain in contact with the first stage after the two stages are uncoupled, i.e., free to separate, measured in terms of areas, weights and pressures.

Where a passage connects the two stages of the rocket for ease in pressurization and the stages are uncoupled substantially at the time of launch (as previously explained), p_1p_2 will be substantially equal and Equation 8 simplifies to:

$$\frac{A_2}{A_1}<2C\left(\frac{W_2}{W_1+W_2}\right) \tag{9}$$

It will be observed that the greater the inequality given by Equations 8 and 9 the more securely the two stages will remain in contact. It will also be noted that by substituting for W_2 in these equations the weight of all additional stages, the equation may be readily applied to multiple stage rockets having more than two stages.

Further, if uncoupling takes place substantially at the time of launching, as has been discussed in connection with the release mechanism illustrated in Figs. 2, 3 and 4, the W_1 and p_1 are the initial values of weight and pressure of the first stage. If uncoupling takes place after launch, W_1 and p_1 can be readily determined since they are functionally related. W_2 and p_2 of course remain at substantially their initial values until after separation occurs.

Turning now to Figs. 5*a*, 5*b*, 6 and 7, we have here illustrated an alternative coupling mechanism which might be used in place of the mechanism illustrated in Figs. 2, 3 and 4. As shown in Fig. 6, the bulkhead **24'** which forms the nose end of the interior chamber of the first stage **10**, has a circular well **70** formed therein to receive the nozzle **28'** of the second stage of the rocket. An opening **24***a'* is provided in the bulkhead and a tube **26'**, only a portion of which is shown in Fig. 6 whose bore registers with the opening **24***a'* is attached thereto; an O-ring (not shown) or other sealing device is secured to the tube **26'**. The tube and the sealing device perform the same functions as described in connection with the previous embodiment.

It will be observed that no flange is provided on the second stage nozzle in this construction. Rather, a slider 72, having the shape shown in Figs. 5*a* and 5*b* is formed thereon. This slider engages the slideway 74 formed in the circular wall of the well 70. For a better understanding of the operation of the slider and slideway, reference should be had to Fig. 7, which represents a vertical section through a circular wall of the well 70 which has been "unrolled" or developed for explanatory purposes. The slideway 74 consists of a tortuous passage formed in the side-wall of the well 70 having substantially the shape shown in Fig. 7.

In use, the nozzle 28' of the second stage 12 of the rocket is inserted in the well 70 with the tube 26' passing into the nozzle orifice 28*a*'. As a result of this insertion, the slider 72 enters the passage 74*a* and follows the passage downwardly until the sloping lower surface 72*a* of the slider engages the lower surface 74*b* of passage 74*a*. The force exerted in putting the two stages together causes the slider to follow the sloping surface 74*b* downwardly until the vertical surface 72*b* of the slider engages the vertical surface 74*c* of the passage. With the two stages in this position pressurization may be started.

As the pressure in the second stage builds up, the second stage tends to push away from the restrained first stage as previously described and the slider moves upwardly along the vertical surface 74*c* until its upper sloping surface 72*c* engages the sloping surface 74*d*. The space between the upper end of surface 74*c* and the surface 74*d* immediately above it is slightly greater than the extent of the surface 72*b*. Thus the pressure will cause the slider 72 to travel up the sloping surface 74*d*, until the slider's vertical surface engages the short vertical surface 74*e*. The second stage is locked in this position (illustrated by the sectional view of the slider in Fig. 7) by the relative pressure force previously mentioned which tends to separate the two stages and the second stage remains in this position during pressurization.

Upon launching of the multiple stage rocket, the second stage will be forced downwardly toward the first stage, as a result of the acceleration as previously explained. As a result of the motion of the second stage, the surface 72*b* of the slider will be disengaged from the surface 74*e* and its lower surface 72*a* will engage the sloping surface 74*f*. The camming action of the two sloping surfaces will cause the slider to move until its surface 72*b* engages the vertical surface 74*g*. The second stage will remain in this position ready for release, while the first stage continues to accelerate. When the thrust of the first stage diminishes enough for the second stage to separate therefrom, the second stage moves upwardly away from the first stage. As it does so, the upper surface 72*c* of the slider engages the sloping surface 74*h* of the passage and causes the slider to enter the substantially vertical passage 74*i*. When the slider is in the passage 74*i*, the second stage rocket is free to separate from the first stage.

In this fashion, the second stage is coupled to the first stage during pressurization, it is uncoupled by the movement of the second stage toward the first stage upon launching of the rocket, and finally separates from the first stage when the first stage thrust has sufficiently diminished. It will be understood, of course, that the construction described herein, the lateral movements of the slider 72 in the passage 74 as described in connection with Fig. 7 will actually be movements along the circumference of a circle, with accompanying rotation of the second stage with respect to the first stage.

It will also be understood that a number of connected passages similar to passage 74 may be formed about the periphery of the well 70 as illustrated in Fig. 7. Thus in the event that the slider 72 is inserted in the passage 74*i*, its lower surface 72*a* will engage the sloping surface 75*a* of the passage 75 and will thereby be properly positioned for the release cycle to take place. If connected passages are formed about the entire prepihery of well 70, the slider may be inserted in any of them. If they do not extend continuously about the periphery, then the exit passage, corresponding to passage 74*i*, of the last connected passage, is marked to indicate that slider 72 should not be inserted therein.

It will thus be seen that we have provided a simple and extremley inexpensive construction for a multiple stage rocket utilizing noncombustible propellants. Rockets made according to our invention have been described particularly with respect to two different embodiments of two-stage rockets, but it will be obvious to those skilled in the art that additional stages may be added utilizing the same constructions.

In our constructions, the second and first stages both consist of hollow vessels having a nozzle at one end thereof. They are secured together by a coupling means which permits a small amount of relative travel between the stages. During pressurization, the force tending to separate the two stages of the rocket holds the second stage at one end of the travel, locking it in position. Upon launching of the rocket, the second stage is forced to the other end of the travel, which releases the coupling means; the second stage is then free to separate from the first stage when the thrust of the first stages diminishes sufficiently so that the relative force between the two stages can overcome the inertial force tending to hold the two stages together.

In order to permit ready pressurization of the two stages, we also provide a narrow passage connecting them. This passage is sufficiently constricted to prevent the flow of an appreciable amount of propellant through said passage from the second stage in the small time required for the first stage to complete its period of acceleration, thus permitting the propellant in the second stage to seal the passage after pressurization.

We have also found that certain relationships must exist between the first and second stage nozzle areas, pressures and weights at the time of uncoupling, if the two stages are to remain in contact as a result of rocket acceleration. When these relationships are observed, the second stage release is determined by the time the entire rocket acceleration diminishes to a value less than that required to hold the second stage to the first.

It will thus be seen that the objects set forth above, among those made apparent from the preceding description, are efficiently attained and, since certain changes may be made in the above constructions without departing from the scope of the invention, it is intended that all matter contained in the above description or shown in the accompanying drawing shall be interpreted as illustrative and not in a limiting sense.

It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described, and all statements of the scope of the invention which, as a matter of language, might be said to fall therebetween.

Having described our invention, what we claim as new and desire to secure by Letters Patent is:

1. A rocket including at least two stages comprising, in combination, a first stage including a storage vessel for pressurized fluid, said vessel having a discharge nozzle for propelling said rocket formed in a portion thereof, a second stage including a storage vessel for pressurized fluid said vessel having a discharge nozzle for propelling said stage formed in a portion thereof, means associated with said rocket for holding said first and second stages in engagement after pressurization, with said first and second stage nozzle not in engagement, means for releasing said holding means, the ratio of the nozzle area of said second stage to the nozzle area of said first stage being selected such that said second stage is held in engagement with said first stage after release of said holding means by the force resulting from acceleration of said first stage.

2. The combination defined in claim 1 in which said rocket is so designed that:

$$\frac{A_2}{A_1}<2C\left(\frac{W_2}{W_1+W_2}\right)\frac{p_1}{p_2}$$

where: A_2 is the nozzle area of said second stage, A_1 is the nozzle area of said first stage, C is the nozzle coefficient of said first stage, W_1 is the weight of said first stage at the time of release of said holding means, W_2 is the weight of said second stage at the time said holding means is released, p_1 is the gage pressure in said first stage at the time of release of said holding means and p_2 is the gage pressure in said second stage at the time said holding means is released.

3. A rocket including at least two stages comprising, in combination, a first stage including a storage vessel for pressurized fluid, said vessel having a discharge nozzle for propelling said rocket formed in a portion thereof, a second stage including a storage vessel for pressurized fluid, said vessel having a discharge nozzle for propelling said second stage formed in a portion thereof, means associated with said rocket for holding said first and second stages in engagement after pressurization, with said first and second stage nozzles not in engagement, a passage interconnecting said first and second stages when said stages are held in engagement, means for releasing said holding means, the ratio of the nozzle area of said second stage to the nozzle area of said first stage being selected such that said second stage is held in engagement with said first stage after release of said holding means by the force resulting from acceleration of said rocket.

4. The combination defined in claim 3 in which said rocket is so designed that:

$$\frac{A_2}{A_1}<2C\left(\frac{W_2}{W_1+W_2}\right)\left(\frac{p_1}{p_2}\right)$$

where: A_2 is the nozzle area of said second stage, A_1 is the nozzle area of said first stage, C is the nozzle coefficient of said first stage, W_1 is the weight of said first stage at the time said holding means is released, W_2 is the weight of said second stage at the time said holding means is released, p_1 is the gage pressure in said first stage at the time of release of said holding means and p_2 is the gage pressure in said second stage at the time said holding means is released.

5. A rocket including at least two stages comprising in combination, a first stage, including a storage vessel, the walls of said vessel defining a hollow interior chamber adapted to enclose fluid under pressure, a discharge nozzle for propelling said stage formed in one part of said chamber for discharge of said fluid, a second stage, said second stage also including a storage vessel, the walls of said vessel defining a second hollow interior chamber adapted to enclose fluid under pressure with a discharge nozzle for propelling said rocket formed in one portion thereof, means for coupling said first stage and said second stage in fixed relative relation with the nozzle of said second stage directed toward a non-nozzle portion of said first stage prior to release of said rocket, means responsive to the acceleration of said rocket after launching thereof to release said coupling means and thereby free said second stage for separation from said first stage.

6. The combination defined in claim 5 which includes a passage interconnecting the storage vessels of said first and second stages when said stages are in engaged relationship.

7. The combination defined in claim 5 in which said coupling means are released by the acceleration of said rocket which occurs substantially at the time said rocket is launched.

8. A rocket including at least two stages comprising in combination, a first stage, including a storage vessel, the walls of said vessel defining a hollow interior chamber adapted to enclose fluid under pressure, a discharge nozzle formed in one part of said chamber for discharge of said fluid to propel said rocket, a second stage, said second stage also including a storage vessel, the walls of said vessel defining a second hollow interior chamber adapted to enclose fluid under pressure with a discharge nozzle for said fluid to propel said second stage formed in one portion thereof, means for coupling said first stage and said second stage in fixed relative relation with the nozzle of said second stage directed toward a non-nozzle portion of said first stage prior to release of said rocket, said coupling means permitting a small travel of said stages with respect to each other, and said coupling means being locked when said second stage is at one end of said travel, the acceleration of said rocket upon launching causing said second stage to move to the other end of said travel to thereby uncouple said second stage and permit said second stage to separate from said first stage.

9. A rocket including at least two stages, comprising, in combination, a first stage including an elongated storage vessel, the walls of said vessel defining a hollow interior chamber adapted to enclose fluid under pressure, a discharge nozzle for propelling said rocket formed in one end of said vessel for discharge of said pressurized fluid, a second stage including an elongated storage vessel with a discharge nozzle for propelling said second stage formed at one end thereof, means for coupling said first stage and said second stage together in end to end relationship, with the nozzle of said second stage engaging the non-nozzle end of said first stage prior to launching said rocket, said coupling means permitting a small longitudinal travel of said stages with respect to each other, said coupling means being locked when said second stage is at the end of said travel most distant from said first stage, the launching of said first stage causing said second stage to move toward said first stage to the other end of said travel, thereby uncoupling said second stage and permitting it to separate from said first stage.

10. The combination defined in claim 9 which includes a passage interconnecting the storage vessels of said first and second stages when said stages are in engaged relationship.

11. A rocket including at least two stages, comprising, in combination, a first stage including an elongated storage vessel, the walls of said vessel defining a hollow interior chamber adapted to enclose fluid under pressure, a discharge nozzle formed in one end of said vessel for discharge of said pressurized fluid, a second stage, said second stage being smaller than said first stage and including an elongated storage vessel with a discharge nozzle formed at one end thereof, a flange formed on the periphery of said nozzle, means for coupling said first stage and said second stage together in end to end relationship, with the nozzle of said second stage engaging the non-nozzle end of said first stage prior to launching said rocket, said coupling means including an arm pivotally attached to said first stage having a hook on one end thereof, said hook being adapted to engage the flange on said second stage nozzle as said arm is rotated about said pivot, resilient means biasing said arm to a position of non-engagement with said flange, and a handle attached to said arm whereby said arm may be rotated about said pivot to engage said flange, said hook being held in engagement with said flange against said resilient means by the relative pressure force tending to separate said first and second stages prior to launch, launching of said first stage causing movement of said first and second stages toward each other, to thereby cause said flange to disengage from said hook, said arm thereby being retracted by said resilient means.

12. The combination defined in claim 11 in which an annular groove is formed on the inner surface of said flange.

13. A rocket including at least two stages comprising, in combination, a first stage including an elongated storage vessel, the walls of said vessel defining a hollow interior chamber adapted to enclose fluid under pressure, a discharge nozzle for propelling said rocket formed in one end of said vessel for discharge of said pressurized fluid, a bulkhead formed across the non-nozzle end of said first stage, an opening formed in said bulkhead, a tubular member secured to the outer surface of said bulkhead, the passage in said tubular member registering with the opening formed in said bulkhead, a second stage, said second stage including an elongated storage vessel with a discharge nozzle formed at one end thereof for propelling said second stage, the inner diameter of said second stage nozzle being slightly greater than the outer diameter of said tubular member, whereby said second stage nozzle may be slid over said tubular member during pressurization, thereby permitting both said first stage and said second stage to be simultaneously pressurized.

14. The combination defined in claim 13 which includes sealing means between said second stage nozzle and the tubular member of said first stage.

15. The combination defined in claim 13 in which an O-ring seal is secured to the outer surface of said tubular member, to thereby seal the nozzle of said second stage and said tubular member.

16. The combination defined in claim 13 in which the diameter of the passage formed in said tubular member is sufficiently small so that liquid in said second stage will not flow therethrough when said second stage is otherwise sealed.

17. A rocket including at least two stages, comprising, in combination, a first stage including an elongated storage vessel, the walls of said vessel defining a hollow

interior chamber adapted to enclose a fluid under pressure, a discharge nozzle formed in one end of said vessel for discharge of said pressurized fluid, a second stage, said second stage including an elongated storage vessel having a discharge nozzle formed at one end thereof, means adapted to couple said first and second stages together in fixed relative relation, said coupling means including a protruding slider formed on one of said stages and a tortuous passage formed in the other of said stages, said passage forming a slideway for said slider when said stages are in engagement, said slideway including an inlet passage for said slider, when said stages are placed in engagement, pressurization of said stages causing said slider to move from said inlet passage to a locked position in said slideway, and the acceleration of said rocket after launching thereof causing said slider to move from said locked position to an outlet passage of said slideway, said first and second stages being free to separate when said slider is in said outlet passage.

18. The combination defined in claim 17 in which a plurality of said tortuous passages are formed on one of said stages, the outlet passage of one slideway being the inlet passage of the next succeeding slideway.

19. The combination defined in claim 17 in which a passage interconnects the storage vessels of said first and second stages when said stages are in engagement.

References Cited in the file of this patent

UNITED STATES PATENTS

2,804,823	Jablansky	Sept. 23, 1957
2,829,491	Teague	Apr. 8, 1958

FOREIGN PATENTS

161,579	Australia	Mar. 1, 1955